

Having learned the natural responses of a second order system both by solving the differential equation in the conventional way and with Laplace transform, let us look at the forced responses using the Laplace Transform technique

1.2.1 Impulse Response of Second Order Systems

Let us consider a second order system represented by the transfer function $G(s) =$

$$\frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$G(s) = \frac{X(s)}{F(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$X(s) = F(s) \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

For a unit impulse $F(t) = \delta(t)$, $F(s) = 1$

$$X(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{\omega_n^2}{(s + \zeta\omega_n)^2 - (\zeta\omega_n)^2 + \omega_n^2} = \frac{\omega_n^2}{[s + \zeta\omega_n]^2 - [\omega_n\sqrt{\zeta^2 - 1}]^2} \dots\dots\dots(24)$$

Critically damped system ($\zeta = 1$)

The equation (24) reduces as follows.

$$X(s) = \frac{\omega_n^2}{(s + \omega_n)^2} = \frac{A}{s + \omega_n} + \frac{B}{(s + \omega_n)^2} \quad \text{(Partial Fraction)}$$

$$\omega_n^2 = A(s + \omega_n) + B \dots\dots\dots(24a)$$

By letting $s = -\omega_n$, we get $B = \omega_n^2$

Equating constants on both sides of (24a), we can write $\omega_n^2 = A\omega_n + B$

We get $A = 0$

$$X(s) = \frac{\omega_n^2}{(s + \omega_n)^2}$$

Taking inverse Laplace Transform

$$x(t) = \omega_n^2 t e^{-\omega_n t} \dots\dots\dots(25)$$

Compare this equation with the natural response obtained for a critically damped second order system with the initial conditions $x(0) = 0$, $\dot{x}(0) = 1$ (#Exercise 6 and 7)

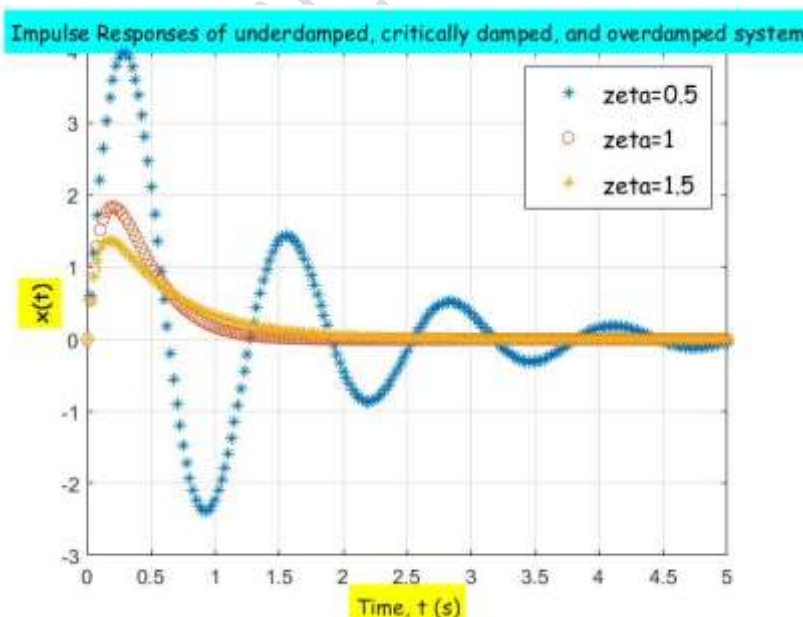


Fig.8 Impulse response of second order system for $\omega_n = 5$ rad/s

Overdamped system ($\zeta > 1$)

The equation (24) is written as (for simplification)

$$X(s) = \frac{\omega_n^2}{(s+a)^2 - b^2} \quad \text{where } a = \zeta\omega_n \text{ and } b = \omega_n\sqrt{\zeta^2 - 1}$$

$$X(s) = \frac{\omega_n^2}{\{(s+a)+b\}\{(s+a)-b\}} = \frac{A}{s+a+b} + \frac{B}{s+a-b}$$

$\omega_n^2 = A(s+a-b) + B(s+a+b)$. Let us put $s=b-a$ and $s=-a-b$ to evaluate A and B

$$\omega_n^2 = A(b-a+a-b) + B(b-a+a+b) \gg B = \frac{\omega_n^2}{2b}$$

$$\omega_n^2 = A(-a-b+a-b) + B(-a-b+a+b) \gg A = -\frac{\omega_n^2}{2b}$$

$$X(s) = \frac{-\frac{\omega_n^2}{2b}}{s+a+b} + \frac{\frac{\omega_n^2}{2b}}{s+a-b} = \frac{\omega_n^2}{2b} \left\{ \frac{1}{s+(a-b)} - \frac{1}{s+(a+b)} \right\}$$

Taking inverse Laplace Transform

$$x(t) = \frac{\omega_n^2}{2b} \{ e^{-(a-b)t} - e^{-(a+b)t} \}$$

$$x(t) = \frac{\omega_n^2}{2\omega_n\sqrt{\zeta^2-1}} \left\{ e^{-(\zeta\omega_n-\omega_n\sqrt{\zeta^2-1})t} - e^{-(\zeta\omega_n+\omega_n\sqrt{\zeta^2-1})t} \right\}$$

$$x(t) = \frac{\omega_n e^{-(\zeta\omega_n)t}}{2\sqrt{\zeta^2-1}} \left\{ e^{(\omega_n\sqrt{\zeta^2-1})t} - e^{-(\omega_n\sqrt{\zeta^2-1})t} \right\}$$

$$x(t) = e^{-\zeta\omega_n t} \left\{ \frac{\omega_n}{2\sqrt{\zeta^2-1}} e^{(\omega_n\sqrt{\zeta^2-1})t} - \frac{\omega_n}{2\sqrt{\zeta^2-1}} e^{-(\omega_n\sqrt{\zeta^2-1})t} \right\} \dots\dots\dots (26)$$

Compare this equation with the natural response obtained for a overdamped second order system with the initial conditions $x(0) = 0$, $\dot{x}(0) = 1$ (#Exercise 6 and 7)

Underdamped system ($\zeta < 1$)

The equation (24) is written as

$$X(s) = \frac{\omega_n^2}{[s+\zeta\omega_n]^2 - [\omega_n\sqrt{\zeta^2-1}]^2} = \frac{\omega_n^2}{[s+\zeta\omega_n]^2 - [j\omega_n\sqrt{1-\zeta^2}]^2} = \frac{\omega_n^2}{[s+\zeta\omega_n]^2 + [\omega_d]^2}$$

Recall that $\omega_d = \omega_n\sqrt{1-\zeta^2}$ and $j^2 = (-1)$

$$X(s) = \left(\frac{\omega_n^2}{\omega_d} \right) \frac{\omega_d}{[s+\zeta\omega_n]^2 + [\omega_d]^2}$$

Taking inverse Laplace Transform (revisit the laplace transforms of $\sin \omega t$)

$$x(t) = \left(\frac{\omega_n^2}{\omega_d} \right) e^{-\zeta\omega_n t} \sin \omega_d t$$

$$x(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin \omega_d t \dots\dots\dots (27)$$

Compare this equation with the natural response obtained for an underdamped second order system with the initial conditions $x(0) = 0$, $\dot{x}(0) = 1$ (#Exercise 6 and 7)

The impulse responses of a second order system with $\omega_n = 5$ rad/s and ζ values of 0.5, 1, and 1.5 are shown in Figure 8. The behavior of the system when an impulse is given as the input (forced response/particular solution) is similar to its natural response (homogeneous solution) with the initial conditions $x(0) = 0$, $\dot{x}(0) = 1$. This is understandable because the impulse acts only for a very short duration initially and thereafter the system behaves naturally as there is no input/forcing term.

#Exercise 10

Given below are the natural frequency and damping ratio of a second order system respectively. Plot the impulse responses using Matlab/Scilab.

- i) 3 rad/s and 0.25 ii) 5 rad/s and 1 iii) 2 rad/s and 1.25 iv) 4 rad/s and 0

#Exercise 11

The transfer functions of second order system are given below. Plot the impulse responses using Matlab/Scilab

i) $G(s) = \frac{24}{3s^2 + 8s + 24}$ ii) $G(s) = \frac{15}{(s+3)(2s+5)}$

#Exercise 12

The poles of three second order systems are given below. Plot the impulse responses using Matlab/Scilab

- i) -1, -5 ii) -3, -3 iii) $-2 \pm j5$

1.2.2 Step Response of Second Order Systems

Consider a second order system with transfer function $G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$

$$G(s) = \frac{X(s)}{F(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$X(s) = F(s) \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

For a unit step input $F(t)=1$, $F(s)=1/s$

$$X(s) = \left(\frac{1}{s}\right) \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \left(\frac{1}{s}\right) \frac{\omega_n^2}{(s+\zeta\omega_n)^2 - (\zeta\omega_n)^2 + \omega_n^2} = \left(\frac{1}{s}\right) \frac{\omega_n^2}{[s+\zeta\omega_n]^2 - [\omega_n\sqrt{\zeta^2-1}]^2} \dots\dots\dots(27a)$$

Critically damped system ($\zeta = 1$)

The equation (27a) becomes

$$X(s) = \left(\frac{1}{s}\right) \frac{\omega_n^2}{(s+\omega_n)^2} = \frac{A}{s} + \frac{B}{s+\omega_n} + \frac{C}{(s+\omega_n)^2} \dots\dots\dots(28)$$

$$\omega_n^2 = A(s+\omega_n)^2 + Bs(s+\omega_n) + Cs$$

Substituting $s=0$, we get $A=1$

When $s = -\omega_n$ $\gggg C = -\omega_n$

Equating coefficients of s^2 , $0 = A+B \gggg B = -1$

$$\text{Therefore } X(s) = \frac{1}{s} - \frac{1}{s+\omega_n} - \frac{\omega_n}{(s+\omega_n)^2}$$

Taking inverse Laplace Transform

$$x(t) = 1 - e^{-\omega_n t} - \omega_n t e^{-\omega_n t}$$

$$x(t) = 1 - (1 + \omega_n t) e^{-\omega_n t} \dots\dots\dots(29)$$

The response curve starts from origin (when $t=0$) and reaches 1 as $t \rightarrow \infty$

#Exercise 13

Plot equation (29) in Matlab/Scilab/Excel for different values of $\omega_n=1, 3$, and 5

Overdamped system ($\zeta > 1$)

We have equation (27)

$$X(s) = \left(\frac{1}{s}\right) \frac{\omega_n^2}{[s + \zeta\omega_n]^2 - [\omega_n\sqrt{\zeta^2 - 1}]^2}$$

$$X(s) = \left(\frac{1}{s}\right) \frac{\omega_n^2}{(s+a)^2 - b^2} \quad \text{where } a = \zeta\omega_n \text{ and } b = \omega_n\sqrt{\zeta^2 - 1}$$

$$X(s) = \frac{\omega_n^2}{s \{(s+a)+b\} \{(s+a)-b\}} = \frac{A}{s+a+b} + \frac{B}{s+a-b} + \frac{C}{s}$$

$$\omega_n^2 = A s(s+a-b) + B s(s+a+b) + C(s+a+b)(s+a-b)$$

Let us put $s=b-a$, $s=-a-b$, and $s=0$ to evaluate B, A, and C respectively.

$$\omega_n^2 = B(b-a)(b-a+a+b) \gg B = -\frac{\omega_n^2}{2b(a-b)}$$

$$\omega_n^2 = A(-a-b)(-a-b+a-b) \gg A = \frac{\omega_n^2}{2b(a+b)}$$

$$\omega_n^2 = C(a+b)(a-b) \gg C = \frac{\omega_n^2}{a^2 - b^2} = 1 \quad (\text{note that } a^2 - b^2 = \omega_n^2)$$

$$X(s) = \left(\frac{\omega_n^2}{2b(a+b)}\right) \frac{1}{s+a+b} - \left(\frac{\omega_n^2}{2b(a-b)}\right) \frac{1}{s+a-b} + \frac{1}{s}$$

Taking inverse Laplace Transform

$$x(t) = \left\{ \omega_n^2 \frac{e^{-(a+b)t}}{2b(a+b)} - \omega_n^2 \frac{e^{-(a-b)t}}{2b(a-b)} + 1 \right\}$$

$$= 1 + \frac{\omega_n^2(a-b)e^{-(a+b)t} - \omega_n^2(a+b)e^{-(a-b)t}}{2b(a^2 - b^2)}$$

$$= 1 + \frac{\omega_n^2(a-b)e^{-(a+b)t} - \omega_n^2(a+b)e^{-(a-b)t}}{2b\omega_n^2}$$

$$= 1 + \frac{(a-b)e^{-(a+b)t} - (a+b)e^{-(a-b)t}}{2b} = 1 + \left(\frac{e^{-at}}{2b}\right) \{(a-b)e^{-bt} - (a+b)e^{bt}\}$$

$$= 1 - \left(\frac{e^{-at}}{2b}\right) \{(a+b)e^{bt} - (a-b)e^{-bt}\}$$

$$x(t) = 1 - \frac{e^{-\zeta\omega_n t}}{2\omega_n\sqrt{\zeta^2 - 1}} \left[(\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}) e^{\omega_n\sqrt{\zeta^2 - 1} t} - (\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}) e^{-\omega_n\sqrt{\zeta^2 - 1} t} \right]$$

$$x(t) = 1 - e^{-\zeta\omega_n t} \left\{ \left(\frac{\zeta + \sqrt{\zeta^2 - 1}}{2\sqrt{\zeta^2 - 1}} \right) e^{\omega_n\sqrt{\zeta^2 - 1} t} - \left(\frac{\zeta - \sqrt{\zeta^2 - 1}}{2\sqrt{\zeta^2 - 1}} \right) e^{-\omega_n\sqrt{\zeta^2 - 1} t} \right\} \dots\dots\dots(30)$$

#Exercise 14

Plot equation (30) in Matlab/Scilab/Excel with $\zeta = 2$ for different values of $\omega_n = 1, 3$, and 5

Underdamped system ($\zeta < 1$)

Let us start from equation (27)

$$X(s) = \left(\frac{1}{s}\right) \frac{\omega_n^2}{[s + \zeta\omega_n]^2 - [\omega_n\sqrt{\zeta^2 - 1}]^2} = \left(\frac{1}{s}\right) \frac{\omega_n^2}{[s + \zeta\omega_n]^2 - [j\omega_n\sqrt{1 - \zeta^2}]^2} = \left(\frac{1}{s}\right) \frac{\omega_n^2}{[s + \zeta\omega_n]^2 + [\omega_d]^2}$$

Recall that $\omega_d = \omega_n\sqrt{1 - \zeta^2}$ and $j^2 = (-1)$

$$X(s) = \left(\frac{1}{s}\right) \frac{\omega_n^2}{[s + \zeta\omega_n]^2 + [\omega_d]^2} = \frac{A}{s} + \frac{Bs + C}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (\text{since the numerator can not be factorised it is kept as such})$$

$$\omega_n^2 = A(s^2 + 2\zeta\omega_n s + \omega_n^2) + (Bs + C)s$$

When $s=0$, $A=1$

$$\text{Equating coefficients of } s^2, \quad 0 = A + B \gg B = -1$$

$$\text{Equating coefficients of } s, \quad 0 = A2\zeta\omega_n + C \gg C = -2\zeta\omega_n$$

$$\begin{aligned}
 X(s) &= \frac{1}{s} - \frac{s+2\zeta\omega_n}{s^2+2\zeta\omega_ns+\omega_n^2} = \frac{1}{s} - \frac{s+2\zeta\omega_n}{[s+\zeta\omega_n]^2 + [\omega_d]^2} \\
 &= \frac{1}{s} - \frac{s+\zeta\omega_n}{[s+\zeta\omega_n]^2 + [\omega_d]^2} - \frac{\zeta\omega_n}{[s+\zeta\omega_n]^2 + [\omega_d]^2} \\
 &= \frac{1}{s} - \frac{s+\zeta\omega_n}{[s+\zeta\omega_n]^2 + [\omega_d]^2} - \left(\frac{\zeta\omega_n}{\omega_d}\right) \frac{\omega_d}{[s+\zeta\omega_n]^2 + [\omega_d]^2}
 \end{aligned}$$

$$X(s) = \frac{1}{s} - \frac{s+\zeta\omega_n}{[s+\zeta\omega_n]^2 + [\omega_d]^2} - \left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right) \frac{\omega_d}{[s+\zeta\omega_n]^2 + [\omega_d]^2}$$

Taking inverse Laplace Transform (revisit the laplace transforms of $\sin \omega t$ and $\cos \omega t$)

$$\begin{aligned}
 x(t) &= 1 - e^{-\zeta\omega_n t} \cos \omega_d t - \left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right) e^{-\zeta\omega_n t} \sin \omega_d t = 1 - e^{-\zeta\omega_n t} \left\{ \cos \omega_d t + \left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right) \sin \omega_d t \right\} \\
 &= 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} \left\{ \sqrt{1-\zeta^2} \cos \omega_d t + \zeta \sin \omega_d t \right\} \\
 &= 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} (X_0 \sin \theta \cos \omega_d t + X_0 \cos \theta \sin \omega_d t) \quad \text{where } X_0 \sin \theta = \sqrt{1-\zeta^2} \text{ and } X_0 \cos \theta = \zeta
 \end{aligned}$$

$$X_0^2 = (1 - \zeta^2) + \zeta^2 = 1 \quad \text{and} \quad \tan \theta = \frac{\sqrt{1-\zeta^2}}{\zeta}$$

$$x(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \theta) \dots\dots\dots(31)$$

#Exercise 15

Plot equation (31) in Matlab/Scilab/Excel with $\zeta = 0.2$ for different values of $\omega_n = 1, 3$, and 5 . Note that the value of θ should be in radian.

The unit step responses of these systems are shown in figure 9.

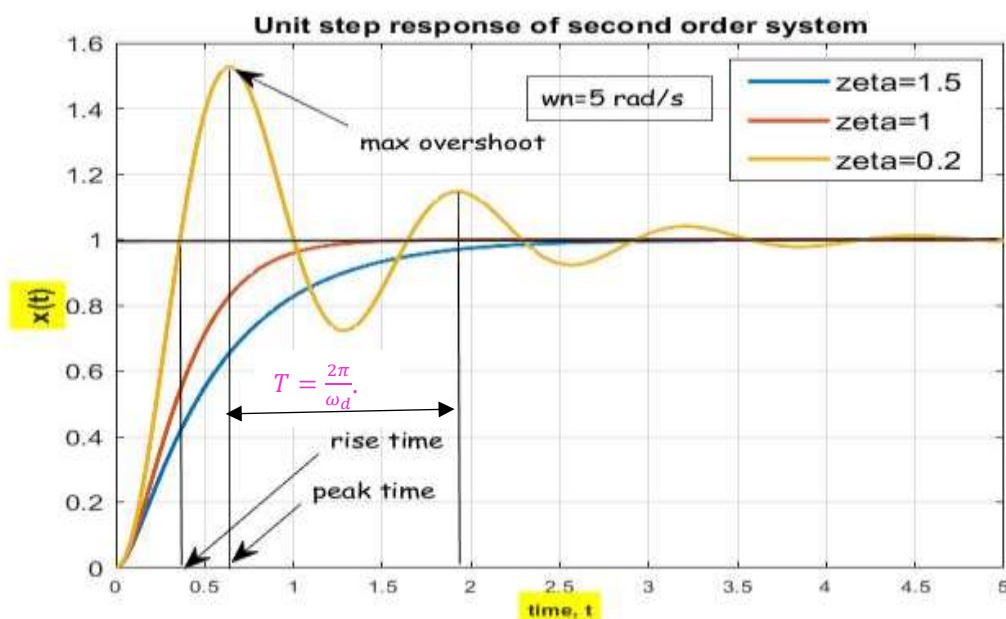


Fig 9. Unit step responses of under, over, and critically damped system for $\omega_n = 5$ rad/s

Looking at the plots of equations (29), (30), and (31), we can understand that the responses of critically damped and overdamped systems reach the desired output without oscillation as $t \rightarrow \infty$, whereas, an underdamped system response will have oscillations before attaining the final value. The frequency (circular frequency) of oscillations is ω_d rad/s with period $T = \frac{2\pi}{\omega_d}$. These oscillations die out after some time and the response reaches a steady state.

#Exercise 16

Using Matlab/Scilab, plot the step responses of a system with $\zeta = 0.1$ for different values of $\omega_n = 1, 3$, and 5 . Repeat it with $\omega_n = 5$ for different values of $\zeta = 0.1, 0.5, 0.8, 1$, and 1.5 . Examine the plots and draw your inferences.